THE BOUNDED BELOW DERIVED CATEGORY OF ADEQUATE MODULES OVER AN AFFINE SCHEME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file adequate.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author's closing 'Faithfulness is proved similarly.' is part of the delivered proof and is quoted verbatim.

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Allowed standard prerequisites (editorial).

- Localisation of a triangulated category at a saturated multiplicative system compatible with the triangulation, and its universal property. Derived Categories, Lemma derived-lemma-triangle-functor-localize, Proposition derived-proposition-construct-localization and Lemma derived-lemma-universal-property-localization, invoked at proof lines 2435-2444. Produces the localisation and the canonical comparison functor.
- Every adequate module embeds into a quasi-coherent sheaf, giving right resolutions inside the quasi-coherent category. Same chapter, Lemma lemma-adequate-characterize and Derived Categories, Lemma derived-lemma-subcategory-right-resolution, invoked at proof lines 2447-2454. Gives essential surjectivity of the comparison functor.

Source reference. Original lemma 05T6: sources/stacks-chow/tags.txt, source lines 7470–7470. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 05R4: sources/stacks-chow/tags.txt, source lines 7398–7398. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 05R7: sources/stacks-chow/tags.txt, source lines 7401–7401. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Source reference. Original proposition 05R6: sources/stacks-chow/tags.txt, source lines 7400–7400. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06VK: sources/stacks-adequate/pinned-original-source.tex, source lines 1787–1787. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $U = \mathrm{Spec}(A)$ be an affine scheme. The bounded below derived category $D^+(Adeq(\mathcal{O}))$ is the localization of $K^+(QCoh(\mathcal{O}_U))$ at the multiplicative subset of universal quasi-isomorphisms.*

Proof. If $\varphi : \mathcal{F}^\bullet \to \mathcal{G}^\bullet$ is a morphism of complexes of quasi-coherent $\mathcal{O}_U$-modules, then $u\varphi : u\mathcal{F}^\bullet \to u\mathcal{G}^\bullet$ is a quasi-isomorphism if and only if φ is a universal quasi-isomorphism. Hence the collection S of universal quasi-isomorphisms is a saturated multiplicative system compatible with the triangulated structure by Derived Categories, Lemma 05R4. Hence $S^{-1}K^+(QCoh(\mathcal{O}_U))$ exists and is a triangulated category, see Derived Categories, Proposition 05R6. We obtain a canonical functor $can : S^{-1}K^+(QCoh(\mathcal{O}_U)) \to D^+(Adeq(\mathcal{O}))$ by Derived Categories, Lemma 05R7.

Note that, almost by definition, every adequate module on U has an embedding into a quasi-coherent sheaf, see Lemma 06VK. Hence by Derived Categories, Lemma 05T6 given $\mathcal{F}^\bullet \in \mathrm{Ob}(K^+(Adeq(\mathcal{O})))$ there exists a quasi-isomorphism $\mathcal{F}^\bullet \to u\mathcal{G}^\bullet$ where $\mathcal{G}^\bullet \in \mathrm{Ob}(K^+(QCoh(\mathcal{O}_U)))$. This proves that can is essentially surjective.

Similarly, suppose that $\mathcal{F}^\bullet$ and $\mathcal{G}^\bullet$ are bounded below complexes of quasi-coherent $\mathcal{O}_U$-modules. A morphism in $D^+(Adeq(\mathcal{O}))$ between these consists of a pair $f : u\mathcal{F}^\bullet \to \mathcal{H}^\bullet$ and $s : u\mathcal{G}^\bullet \to \mathcal{H}^\bullet$ where s is a quasi-isomorphism. Pick a quasi-isomorphism $s' : \mathcal{H}^\bullet \to u\mathcal{E}^\bullet$. Then we see that $s' \circ f : \mathcal{F} \to \mathcal{E}^\bullet$ and the universal quasi-isomorphism $s' \circ s : \mathcal{G}^\bullet \to \mathcal{E}^\bullet$ give a morphism in $S^{-1}K^+(QCoh(\mathcal{O}_U))$ mapping to the given morphism. This proves the "fully" part of full faithfulness. Faithfulness is proved similarly. □